SPREADOUT OF ETALENESS FROM THE CLOSED SET DEFINED BY AN IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The set of points where a ring map is not etale is closed, and is the zero set of an ideal. Algebra, Definition \ref{algebra-definition-etale}, cited at proof lines 1829-1831. Identifies the failure locus with the zero set of an ideal.
- Two closed subsets with empty intersection are defined by ideals generating the unit ideal. Algebra, Lemma \ref{algebra-lemma-Zariski-topology}, cited at proof lines 1833-1834. Gives the decomposition of one used to choose the element.

Source reference. Original lemma 07M0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9710–9710. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00U1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1095–1095. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00E0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 517–517. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Lemma 0.2. *Let $A \to B$ be a ring map and $J \subset B$ an ideal. If $A \to B$ is étale at every prime of $V(J)$, then there exists a $g \in B$ mapping to an invertible element of B/J such that $A' = B_g$ is étale over A.*

Proof. The set of points of $\mathrm{Spec}(B)$ where $A \to B$ is not étale is a closed subset of $\mathrm{Spec}(B)$, see Algebra, Definition 00U1. Write this as $V(J')$ for some ideal $J' \subset B$. Then $V(J') \cap V(J) = \emptyset$ hence $J + J' = B$ by Algebra, Lemma 00E0. Write $1 = f + g$ with $f \in J$ and $g \in J'$. Then g works. □